

Chapter 2

Abstract

Extended Scholarly Abstract

The Theory of Entropicity (ToE) is a theoretical framework that seeks to establish entropy as the primary organizing principle underlying spacetime geometry, gravitation, and physical law. The theory is motivated by a growing body of evidence indicating deep connections between information, thermodynamics, quantum theory, and gravity. Rather than treating entropy as a derived statistical quantity emerging from microscopic physical processes, ToE proposes that entropy constitutes a fundamental dynamical substrate from which information structures, geometric relations, and gravitational phenomena emerge.

The mathematical framework of the theory is constructed upon a generalized manifold architecture known as the Hybrid Manifold Architecture System (HMAS). This manifold contains three interconnected geometric sectors: the Fisher–Rao sector representing classical information geometry, the Fubini–Study sector representing quantum information geometry, and the Lorentzian sector representing emergent spacetime geometry. Each sector possesses an independent metric structure while remaining dynamically connected through an entropy scalar field and a generalized action principle.

To accommodate the simultaneous treatment of multiple geometric sectors, ToE introduces a hierarchical tensor formalism known as the Obidi Convention. Within this framework, conventional tensor indices are supplemented by secondary sector labels, producing a generalized tensor calculus capable of describing interactions among informational and geometric domains. Associated algebraic rules collectively known as Obidi Calculus govern addition, multiplication, contraction, and summation operations across sectors.

The dynamical core of the theory is provided by the Local Obidi Action (LOA), an entropy-coupled generalization of the Einstein–Hilbert action. Variation of the LOA yields the Obidi Field Equations, which govern the evolution of the entropy field and the geometry of each sector. These equations constitute the master dynamical system of the Theory of Entropicity.

A central result of the theory is that Einstein's Field Equations emerge naturally in the low-entropy-gradient limit. When entropy gradients vanish or become sufficiently small, the informational sectors decouple, leaving only the Lorentzian sector dynamically active. In this regime the Obidi Field Equations reduce precisely to

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu},$$

thereby reproducing General Relativity as an emergent approximation of a more general entropic framework.

The Theory of Entropicity therefore extends previous thermodynamic and information-theoretic approaches to gravity by explicitly incorporating classical information geometry, quantum information geometry, and spacetime geometry into a unified field-theoretic structure. The framework provides a systematic

mathematical language for investigating the possibility that gravity, geometry, and physical law are emergent manifestations of deeper entropic processes.

Chapter 3

Introduction and Motivation

3.1 Historical Background

The history of theoretical physics may be understood as a sequence of progressively deeper descriptions of nature.

Classical mechanics interpreted physical phenomena in terms of forces acting within an absolute framework of space and time. Newtonian gravitation described attraction through a universal inverse-square law, providing extraordinary predictive success but offering no explanation for the mechanism by which gravitational influence propagated.

The twentieth century witnessed a profound conceptual transformation through the development of General Relativity. Einstein replaced the notion of gravitational force with spacetime geometry. In this new framework, matter and energy determine the curvature of spacetime, while spacetime curvature governs the motion of matter.

The Einstein Field Equations,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu},$$

represent one of the most successful theories in the history of science.

Yet despite their remarkable predictive power, these equations leave unanswered a fundamental question.

Why should geometry possess dynamical properties at all?

This question becomes increasingly important when one considers the growing evidence linking gravitation with thermodynamics and information.

3.2 The Thermodynamic Revolution in Gravity

The modern relationship between gravity and entropy emerged from investigations of black holes.

In the early 1970s, Jacob Bekenstein proposed that black holes possess entropy proportional to the area of their event horizons. Shortly thereafter, Stephen Hawking demonstrated that black holes emit thermal radiation with a well-defined temperature.

Together these discoveries led to the celebrated Bekenstein–Hawking entropy relation,

$$S_{BH} = \frac{k_B A}{4\ell_P^2},$$

where

$$\ell_P^2 = \frac{G\hbar}{c^3}.$$

This result revealed an unexpected connection between geometry, quantum theory, and thermodynamics.

A geometric object—an event horizon—possesses entropy.

A gravitational system possesses temperature.

A spacetime structure behaves thermodynamically.

These observations suggested that gravitation may not be fundamentally geometric but may instead arise from deeper statistical or informational principles.

3.3 Jacobson's Thermodynamic Derivation

A major conceptual advance occurred in 1995 when Ted Jacobson demonstrated that Einstein's equations can be derived from thermodynamic considerations.

Jacobson considered local Rindler horizons associated with uniformly accelerated observers and imposed the Clausius relation

$$\delta Q = T dS.$$

Remarkably, requiring this relation to hold for all local horizons yields the Einstein Field Equations.

The significance of this result extends far beyond the derivation itself.

If Einstein's equations can be derived from thermodynamic principles, then gravitation may be analogous to pressure, temperature, or elasticity—an emergent macroscopic phenomenon rather than a fundamental interaction.

This observation constitutes one of the intellectual foundations upon which the Theory of Entropicity is built.

3.4 Entropic Gravity and Verlinde's Proposal

A further development occurred through the work of Erik Verlinde.

Verlinde proposed that gravity is not a fundamental force but an emergent entropic phenomenon arising from information associated with holographic screens.

In this framework, gravitational attraction results from entropy gradients.

Matter alters informational configurations, generating effective forces that reproduce Newtonian gravitation and, under appropriate conditions, Einsteinian gravity.

Although controversial, Verlinde's proposal reinforced a growing realization within theoretical physics.

Entropy is not merely a thermodynamic quantity.

Entropy appears intimately connected to the structure of spacetime itself.

3.5 Information Geometry as a Physical Structure

At approximately the same time, developments in information geometry revealed that probability distributions possess intrinsic geometric structure.

For a family of distributions $p(x|\theta)$, the Fisher information metric

$$g_{ij} = \int p(x|\theta) \partial_i \ln p \partial_j \ln p dx$$

defines a Riemannian geometry on the space of statistical states.

Similarly, quantum states possess a natural geometric structure described by the Fubini–Study metric.

These developments established a profound principle.

Information is not merely abstract.

Information possesses geometry.

Once this fact is recognized, a conceptual chain emerges:

$$\text{Information} \longrightarrow \text{Geometry}.$$

Because General Relativity identifies gravity with geometry, one may continue this chain:

$$\text{Information} \longrightarrow \text{Geometry} \longrightarrow \text{Gravity}.$$

This observation becomes a crucial element in the conceptual architecture of the Theory of Entropicity.

3.6 The Fundamental Question of ToE

The preceding developments collectively suggest an important question.

Why do entropy and information repeatedly appear at the deepest known boundaries of physics?

Entropy appears in:

- Statistical mechanics.
- Thermodynamics.
- Black-hole physics.
- Quantum theory.
- Information theory.
- Cosmology.
- Gravitational thermodynamics.

Information appears in:

- Quantum measurement.
- Statistical inference.
- Holography.
- Entanglement.
- Information geometry.

The recurrence of these concepts across otherwise distinct domains raises the possibility that they are not secondary features of physical reality.

They may instead be indicators of a deeper substrate.

The Theory of Entropicity begins with this possibility.

3.7 The Foundational Hypothesis

The foundational hypothesis of ToE may be stated succinctly:

Entropy is the primary dynamical substrate from which information, geometry, gravitation, and physical law emerge.

This hypothesis reverses the conventional explanatory hierarchy.

Traditional approaches generally assume

Matter $\longrightarrow$ Information $\longrightarrow$ Entropy.

The Theory of Entropicity proposes instead

Entropy $\longrightarrow$ Information $\longrightarrow$ Geometry $\longrightarrow$ Gravity $\longrightarrow$ Matter Dynamics.

The theory therefore treats entropy not as a statistical descriptor of physical systems but as a fundamental constituent of physical reality.

3.8 Motivation for the Hybrid Manifold

If entropy influences both classical and quantum informational structures, then a purely Lorentzian spacetime manifold becomes insufficient.

The geometry of information itself must be incorporated into the mathematical framework.

This requirement motivates the introduction of the Hybrid Manifold Architecture System.

Within HMAS, the Fisher–Rao sector captures classical information geometry, the Fubini–Study sector captures quantum information geometry, and the Lorentzian sector captures emergent spacetime geometry.

Together these sectors provide a mathematical environment in which informational and gravitational phenomena may be studied simultaneously.

3.9 Motivation for the Obidi Convention

A manifold containing multiple geometric sectors requires a corresponding extension of tensor notation.

Standard tensor calculus is designed for a single geometric structure.

ToE introduces the Obidi Convention to generalize tensor operations to the multi-sector setting.

The convention allows all physical quantities to carry both geometric and informational identities simultaneously.

This extension is not merely a notational convenience.

It forms the mathematical foundation upon which the field equations of the theory are constructed.

3.10 Objectives of the Present Work

The principal objectives of this monograph are:

1. To establish the mathematical foundations of the Theory of Entropicity.
2. To define rigorously the Hybrid Manifold Architecture System.
3. To formulate the Obidi Convention and Obidi Calculus.
4. To derive the Local Obidi Action.
5. To obtain the Obidi Field Equations.
6. To demonstrate the emergence of Einstein gravity as a limiting case.
7. To place ToE within the broader context of modern information-theoretic approaches to gravitation.

The chapters that follow develop each of these objectives systematically and rigorously, beginning with the mathematical foundations required for the construction of the theory.